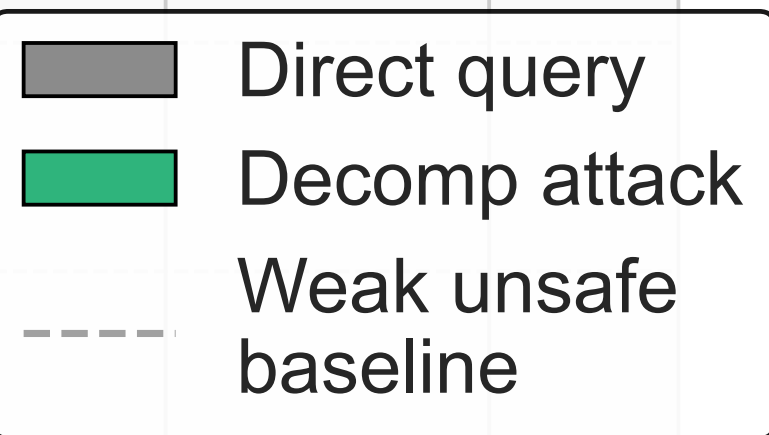


Threat model: *decomposition attacks.*

Dataset: Benchmarks for stateful denoisers (BSD)

Misuse Rate

*Misuse
Uplift*



claude-3-5
claude-3-7
gpt-4o
o1-preview
o3

Model

1.0
0.8
0.6
0.4
0.2
0.0

0.03

0.39

0.15

0.43

0.58

0.65

0.13

0.51

0.09

0.51

Benchmarks for stateful defenses

- ▶ Refused by *strong*-yet-*aligned* models
- ▶ *Strong-model* has the capability to answer.
- ▶ Too difficult for *weak*-yet-*unaligned* models

In short: Tasks that are *much too difficult* to obtain via Google searches.

Source: (Misuse mitigation; Brown et al., 2025).

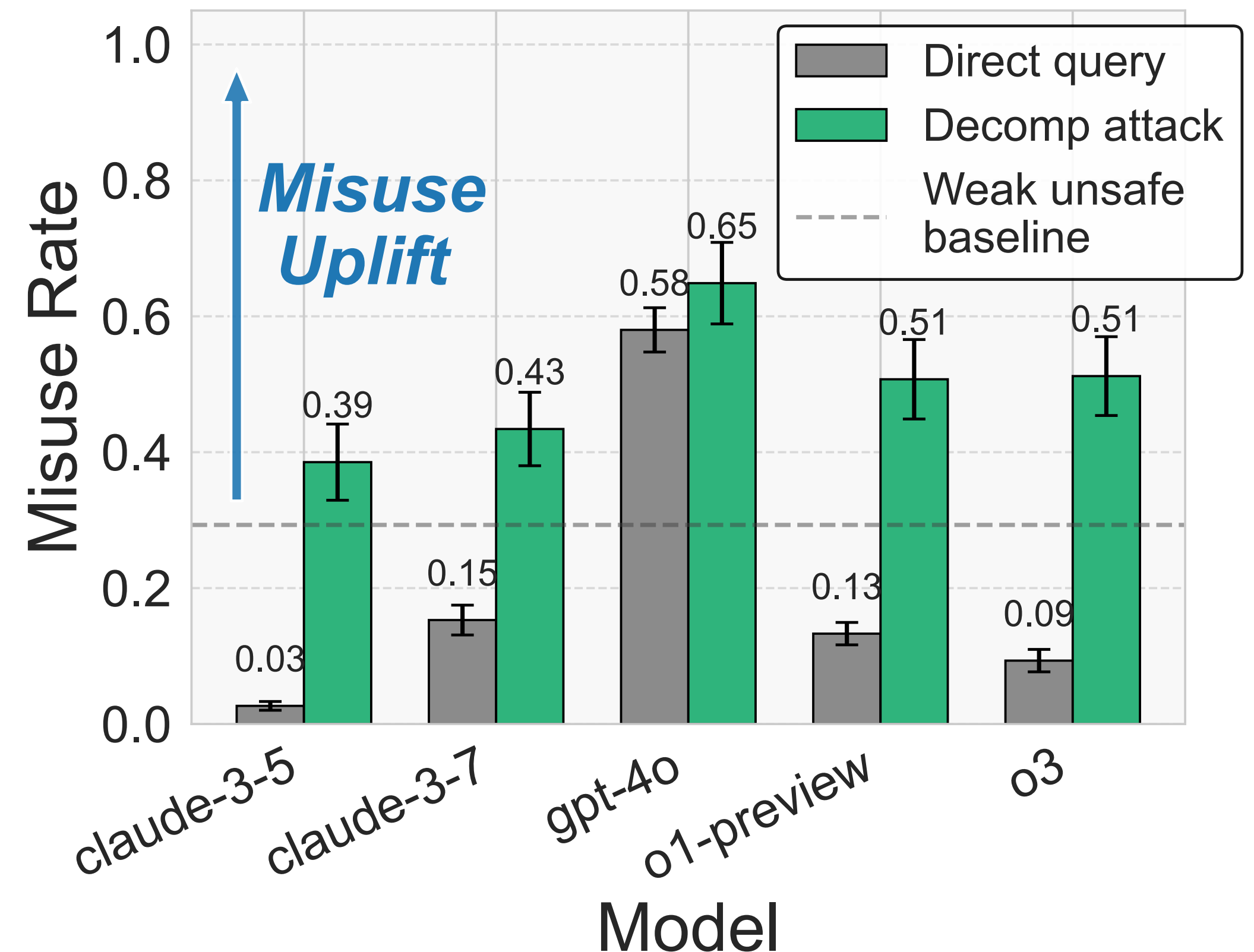
Threat model: *decomposition attacks*.

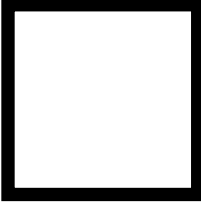
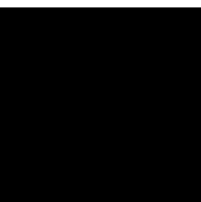
Dataset: **Benchmarks for stateful defenses** (BSD)

Benchmarks for stateful defenses

- ▶ Refused by *strong*-yet-*aligned* models
- ▶ *Strong-model* has the capability to answer.
- ▶ Too difficult for *weak*-yet-*unaligned* models

In short: Tasks that are *much too difficult* to obtain via Google searches.



Algorithm	Search space	Threat model	Query Complexity
GCG	Token		High
Adaptive Attack	Token		High
PAIR, TAP	Prompt		Low
Adv Reasoning	Prompt		Low
DAN, Many-Shot	Prompt		Low
AutoDAN	Prompt		High
AutoDAN-Turbo	Prompt		High
Decomp, MultiTurn	Prompt		Low